EIE328 EXPERIMENT 9.docx
Document

Please can you just break this down
And answer the questions as well

Absolutely. This experiment is really about **two things**:

1. **Binary numbers and BCD**
2. **Logic gates (NOT, AND, OR, NAND, NOR)**

I'll break it down and answer the questions.

PART 1: NUMBER SYSTEMS

Key Idea

In any number system:

$$\text{Value} = \text{digit} \times (\text{radix})^{\text{column number}}$$

Where:

- Radix = base of the number system
- Decimal → radix 10
- Binary → radix 2
- Quinary → radix 5

Question 1

What decimal value has a digit 3 in column No. 2 in a quinary system?

Quinary means radix = 5.

$$3 \times 5^2$$

$$3 \times 25$$

$$= 75$$

☑ **Answer: 75**

Question 2

What decimal value has digit 1 in column No. 5 of a binary system?

Binary radix = 2

$$1 \times 2^5$$

$$= 32$$

☑ **Answer: 32**

Practical 1

A lamp ON = Binary 1

A lamp OFF = Binary 0

Switch UP = 1

Switch DOWN = 0

Question 1.3

(a) Slide UP

Lamp = ON

(b) Slide DOWN

Lamp = OFF

☑ Answers:

Slide	Lamp
Up	ON
Down	OFF

Exercise 1

The table becomes:

Binary State	Lamp	Slide
0	OFF	DOWN

Binary State	Lamp	Slide
1	ON	UP

Question 4

> How many binary columns are necessary to represent numbers up to 9?

Let's count:

Bits	Maximum Number
1	1
2	3
3	7
4	15

To reach 9 we need 4 bits.

- [x] **Answer: 4 bits**

Practical 2 Binary Equivalents

Decimal	Binary
0	0000

Decimal	Binary
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

Question 5

How many different numbers can four binary digits represent?

$$2^4 = 16$$

☑ **Answer: 16 numbers**

(0 to 15)

BCD

BCD represents each decimal digit separately using 4 bits.

Example:

26

becomes

2 = 0010

6 = 0110

Therefore:

26 = 0010 0110

Question 6

How many bits are needed to represent 99 in pure binary?

99 in binary:

$$99_{10} = 1100011_2$$

Count bits:

7 bits

☑ **Answer: 7 bits**

Question 7

Is pure binary more or less efficient than BCD?

BCD for 99:

```
1001 1001
```

= 8 bits

Pure binary:

```
1100011
```

= 7 bits

Uses fewer bits.

☑ **Answer: Pure binary is more efficient**

Question 8

Is there any restriction on BCD bit patterns?

Yes.

BCD only represents decimal digits 0-9.

Valid codes:

```
0000 to 1001
```

Invalid:

```
1010
1011
1100
1101
1110
1111
```

✅ **Answer:** Only 0000 to 1001 are allowed.

PART 2: LOGIC GATES

NOT Gate

NOT flips the input.

A	A'
0	1
1	0

NOT Truth Table

Input	Output
0	1
1	0

Question 1

> If A' = NOT A, what is A'' ?

Apply NOT twice.

Example:

A = 1

A' = 0

A'' = 1

Example:

A = 0

A' = 1

A'' = 0

So:

$$A'' = A$$

☑ **Answer: A'' = A**

NAND Gate

Definition:

$$Z = (AB)'$$

Read as:

"NOT (A AND B)"

NAND Truth Table

A	B	Z
0	0	1
0	1	1
1	0	1

A	B	Z
1	1	0

AND Gate

Definition:

$$Z = AB$$

Both inputs must be 1.

AND Truth Table

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

AND Identities

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

OR Gate

Definition:

$$Z = A + B$$

At least one input must be 1.

OR Truth Table

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

Question 2

How does OR differ from ordinary addition?

Ordinary addition:

A	**B**	**A+B**
0	0	0
0	1	1
1	0	1
1	1	2

OR operation:

A	**B**	**A OR B**
0	0	0
0	1	1
1	0	1
1	1	1

Difference:

For

$$1 + 1$$

Arithmetic gives:

$$2$$

OR gives:

$$1$$

☑ **Answer:** OR does not produce 2 when both inputs are 1.

Exercise 2

Using De Morgan:

$$(A'B')' = A + B$$

Truth table:

A	B	A'	B'	A'B'	(A'B')'	A+B
0	0	1	1	1	0	0
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	0	1	1

Last two columns are identical.

De Morgan's Theorem

The two forms to memorize are:

$$(AB)' = A' + B'$$

and

$$(A + B)' = A'B'$$

These are among the most important formulas in digital logic.

Exercise 3

(a)

$$(AB)' = A' + B'$$

(b)

$$(A + B)' = A'B'$$

(c)

$$(A + B + C)' = A'B'C'$$

(d)

$$(ABC)' = A' + B' + C'$$

NOR Gate

Definition:

$$Z = (A + B)'$$

Read as:

"NOT OR"

NOR Truth Table

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

Exercise 5

From the OR truth table:

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

- [x] These are the four OR identities.

Quick Exam Memory Sheet

NOT

$$A' = \text{NOT A}$$

AND

$$AB$$

Only 1 when both inputs are 1.

OR

$$A + B$$

1 when either input is 1.

NAND

$$(AB)'$$

AND followed by NOT.

NOR

$$(A+B)'$$

OR followed by NOT.

De Morgan

$$(AB)' = A' + B'$$

$$(A+B)' = A'B'$$

If you're preparing this for a lab report or practical test, the **most likely questions** are the binary conversions, the five truth tables (NOT, AND, OR, NAND, NOR), and De Morgan's theorem. Those three topics carry almost the entire experiment.